

Modulation:-

is a technique that changes the characteristics of carrier freq. in according with input signal.

Need of Modulation:-

1- To avoid the interference.

أي أنه كل الموجات قبل عملية ال Modulation تكون

متداخلة فيما بينها فيحصل تداخل بين الموجات

لذلك تتم عملية Modulation للموجات بوضع نظم عملية

ال Modulation للموجات كإشارة تصل إلى مسافات بعيدة

كل إشارة للموجة تعديل كإشارة
ما يحصل تداخل بينه وبين موجة
أخرى

(من حيث التردد)

2) To get desired antenna height.

Which mean that ... the antenna height is directly proportional to the λ ... and λ inversely proportional with Freq.

So Antenna height also inversely proportion with Freq.

3- The ability of Multiplexing.

يعني مثلاً أنا أشتري إرسال موجة تردد

10 kHz ... يعني كل شيء يبدأ من التردد يكون

التردد ... يعني كل شيء يرتفع بشكل كبير وعنا، أرضنا

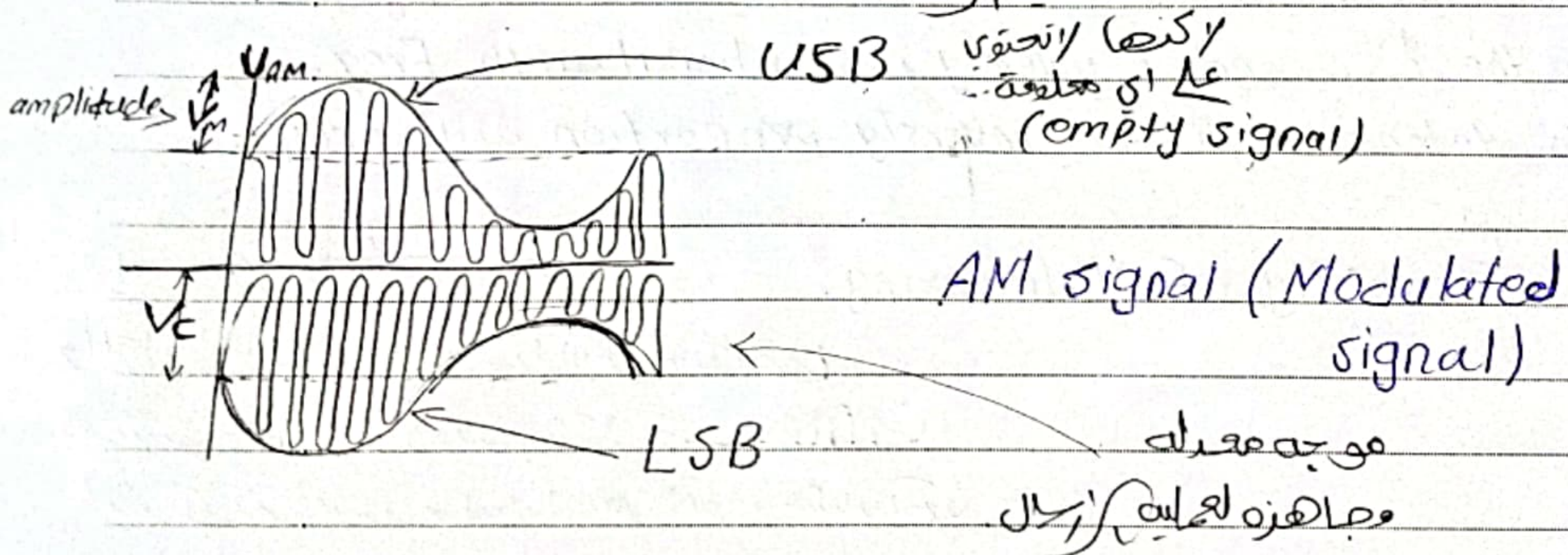
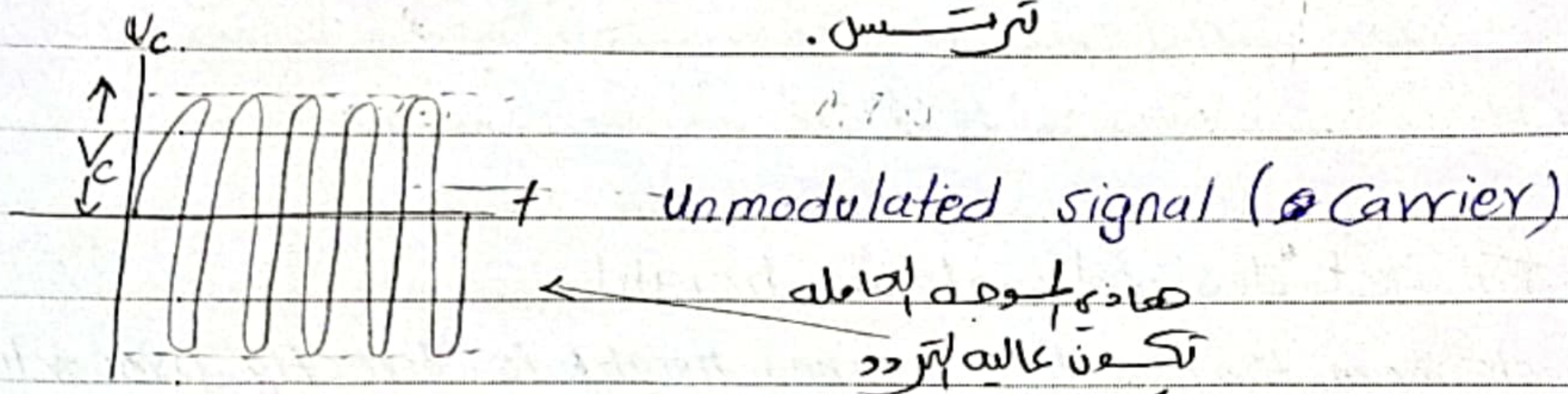
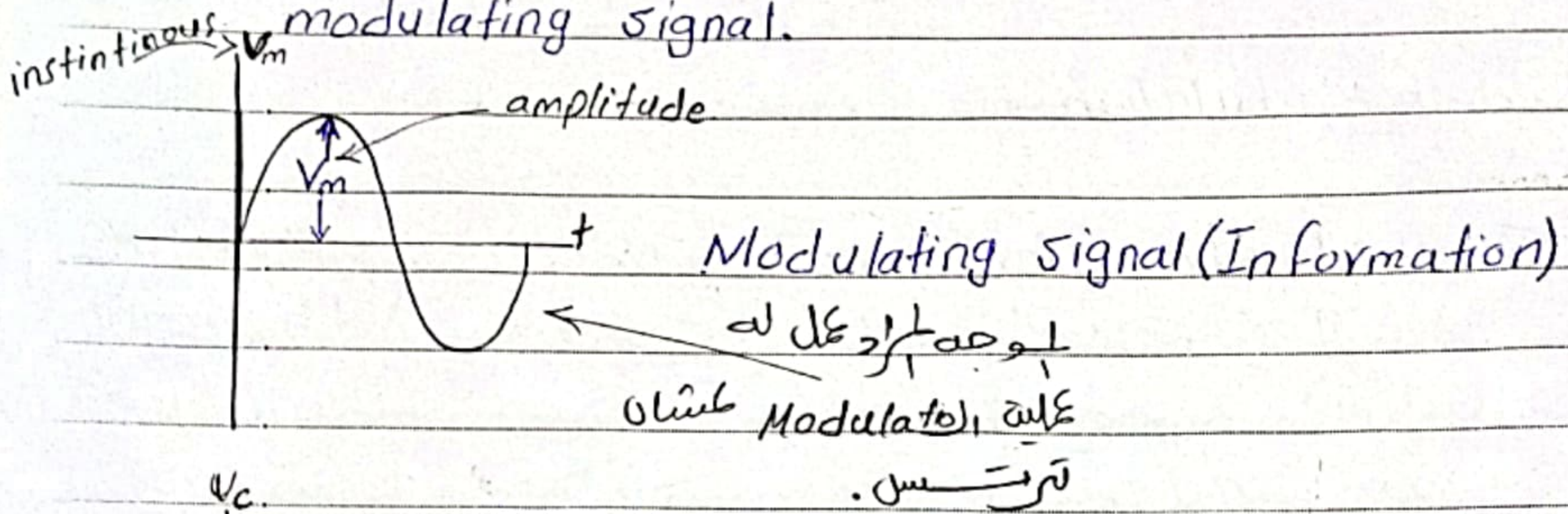
كإشارة كذا اسمها Modulation للموجات كإشارة تكون

تردد ... إلى ... إلى الارتفاع الإشارة يكون بشكل منطقي

Follow.

- AM modulation.

In AM, the amplitude of the carrier is change due to modulating signal.



Follow.

amplitud.
(Peak)

$$V_m = V_m \sin \omega_m t$$

carrier signal.

$$V_c = V_c \sin \omega_c t$$

والموجة

and

$$V_{AM} = V_c + V_m \sin \omega_m t$$

$$V_{AM} = V_{AM} \sin \omega_c t \Rightarrow V_{AM} = (V_c + V_m \sin \omega_m t) \sin \omega_c t$$

الموجة

$$V_{AM} = V_c \sin \omega_c t + V_m (\sin \omega_m t \sin \omega_c t) \Rightarrow = \frac{1}{2} (\cos(A-B) - \cos(A+B))$$

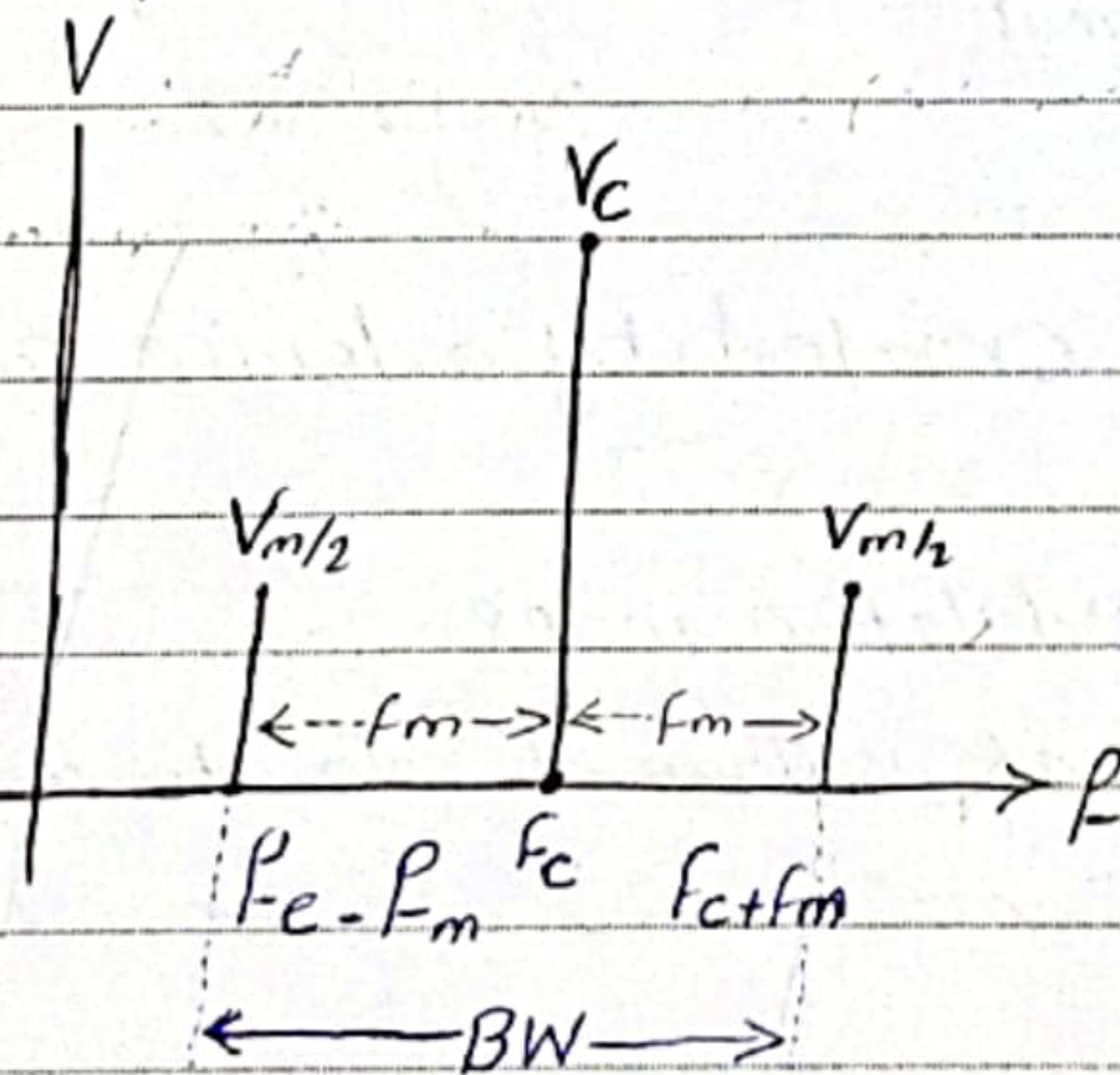
$$V_{AM} = V_c \sin \omega_c t - \frac{V_m}{2} \cos(\omega_c + \omega_m) + \frac{V_m}{2} \cos(\omega_c - \omega_m)$$

Carrier

USB

LSB

Frequency spectrum:



$$BW = 2f_m$$

Modulator:-

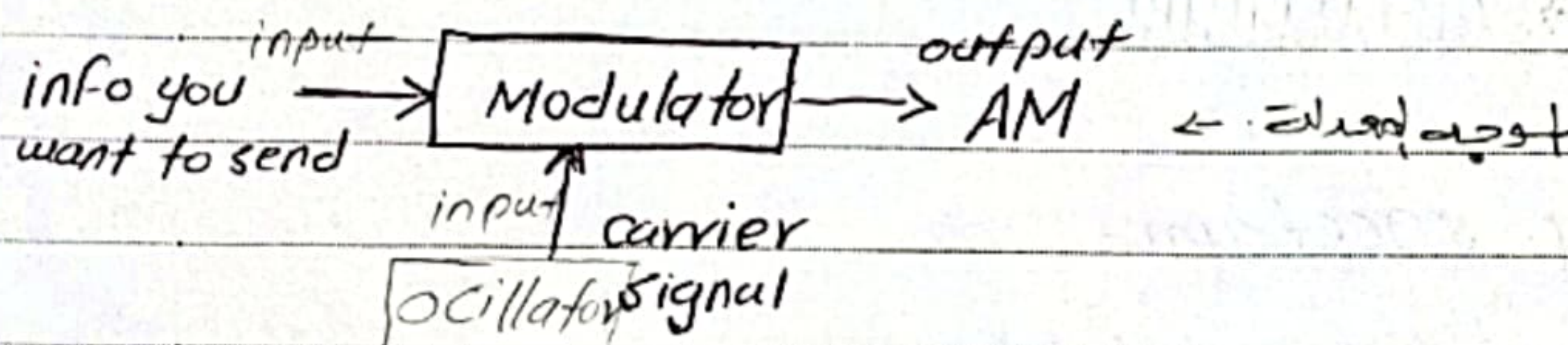
is a circuit that change the low frequency to high frequency.
it's used in transmitting system.

Demodulator (Detector):-

is a circuit used to recover the original information. it's used in a receiver system.

Note:-

Modulator has two input.. one for the info. you want to send it.. and the other for carrier.. also has one output for the AM wave.



Note:-

the carrier signal is created by a device called oscillator

Modulation index:-

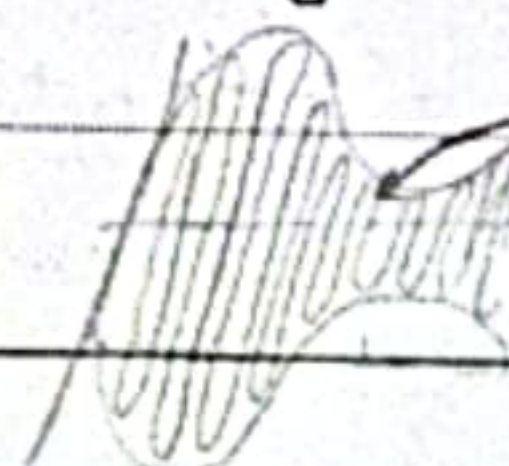
is the ratio of the peak voltage of info.. to the peak voltage of carrier: $m = \frac{V_m}{V_c}$

Note:-

the purpose of modulation index is that to know,

How depth the info. signal in the carrier

يعني يعرفنا كم عمق
الرسالة في الناقل



this is the depth
that i mean

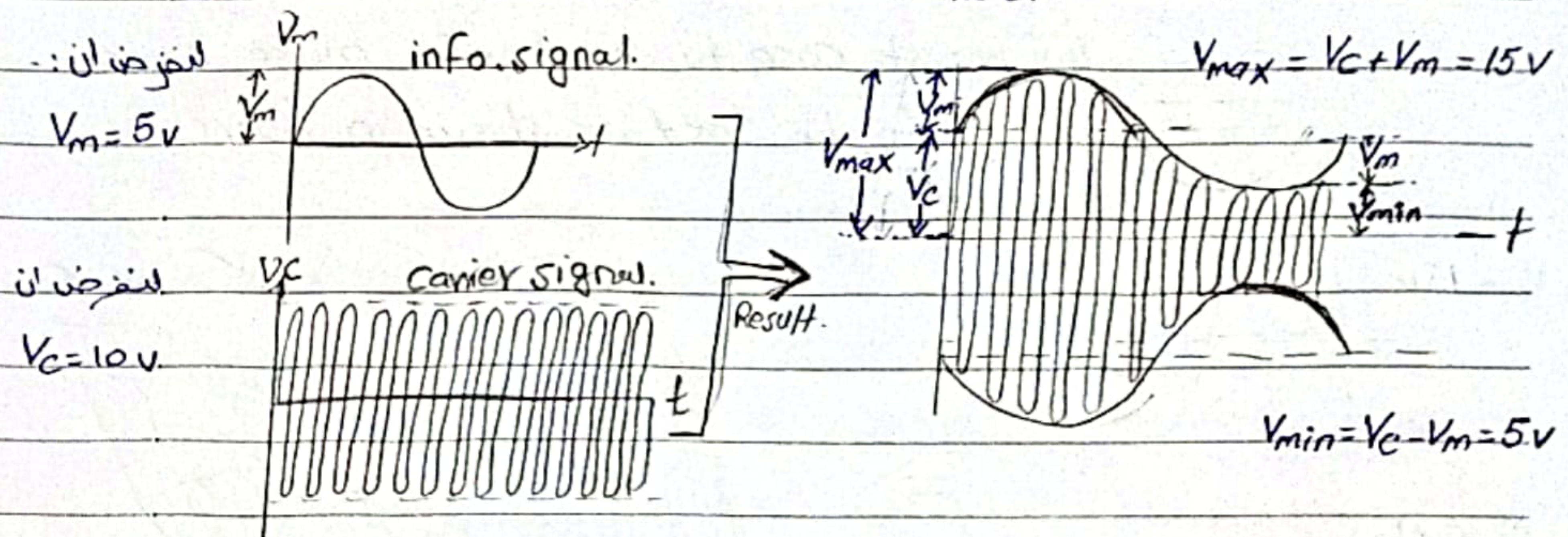
هذا هو العمق
الذي أعني

Follow

There are three cases of M. Index:-

First case:-

$V_m < V_c$ When the amplitude of signal is less than the amplitude of carrier signal.
 $m < 1$ This case called "Under modulation".

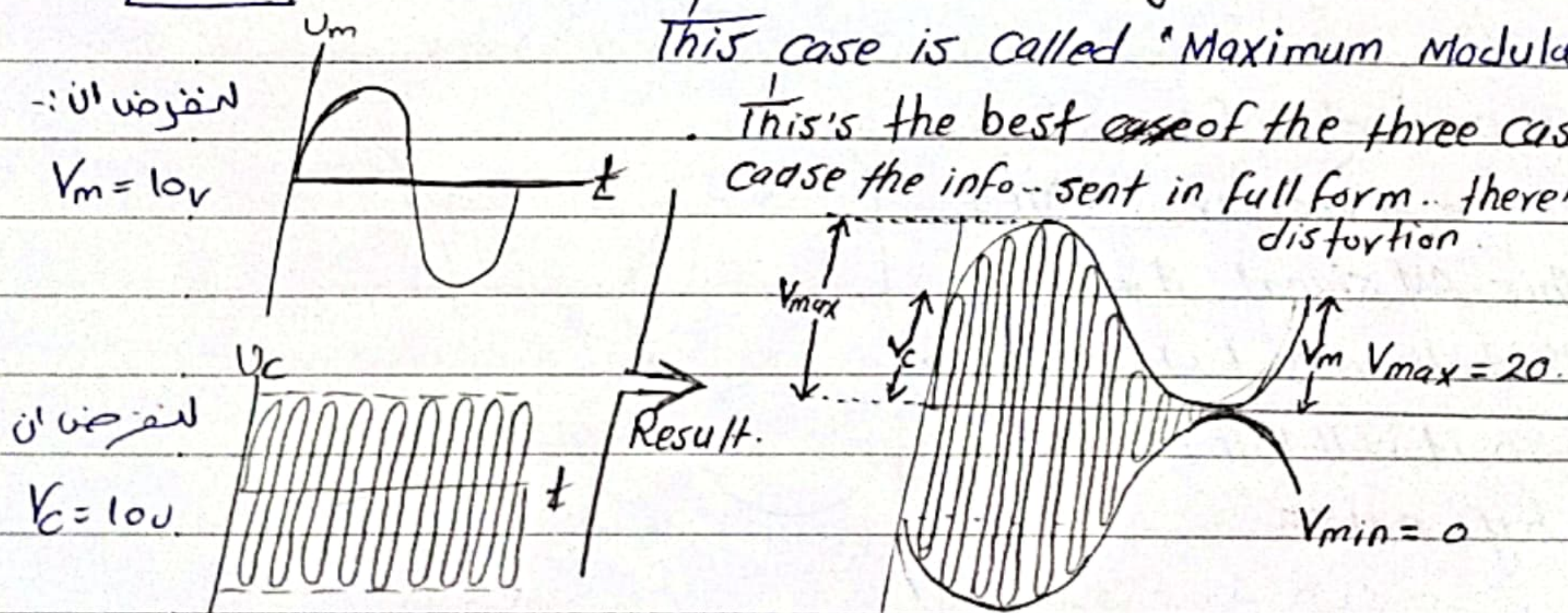


second case :-

$V_m = V_c$ When the amp. of signal is equal to the amp. of carrier signal.
 $m = 1$

This case is called "Maximum Modulation".

This is the best case of the three case cause the info. sent in full form.. there's no distortion.



بني الحالة نلاحظ ان ال amp تبع ال 2 سيجل متساو
 و هذي الحالة تعتبر افضل نوع من المعالين التارن لئن لسيجل تنتقل بشكل سليم

Follow.

The final 3rd case:-

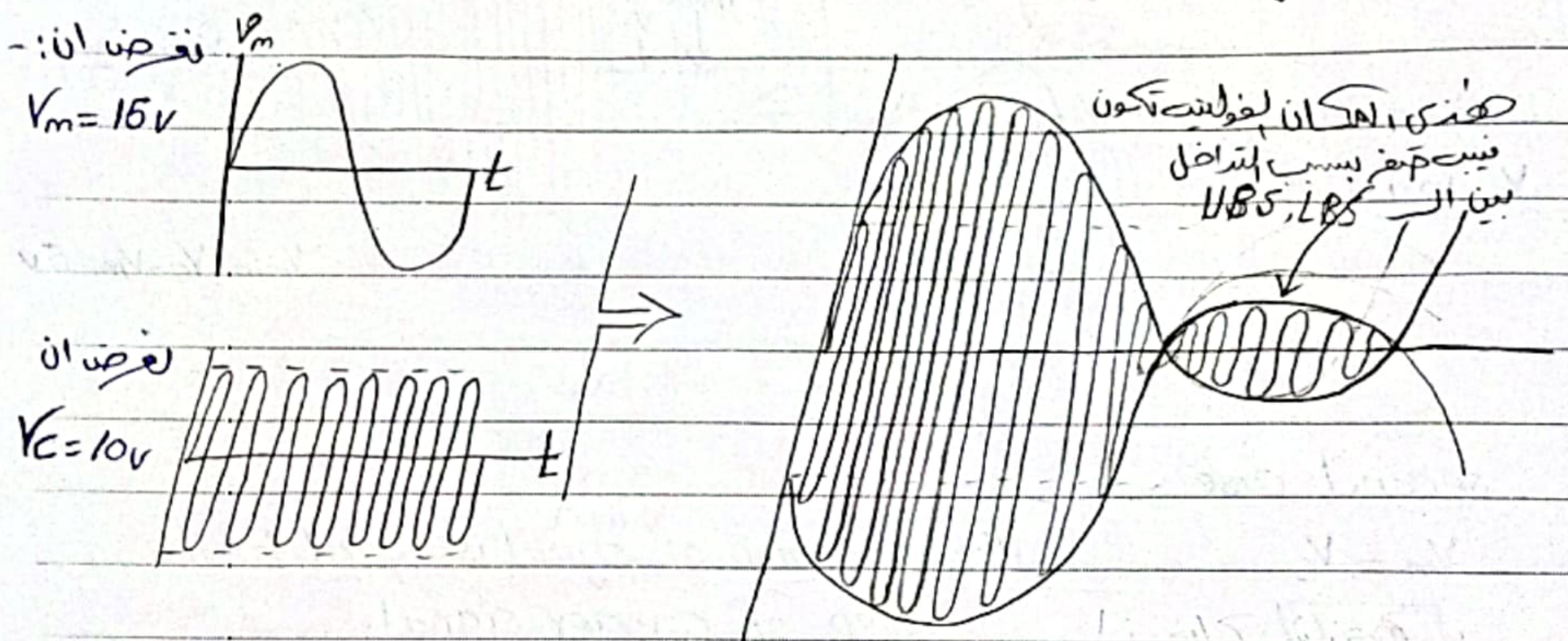
$$V_m > V_c$$

$$m > 1$$

When the amp. of signal is greater than the amp. of carrier.

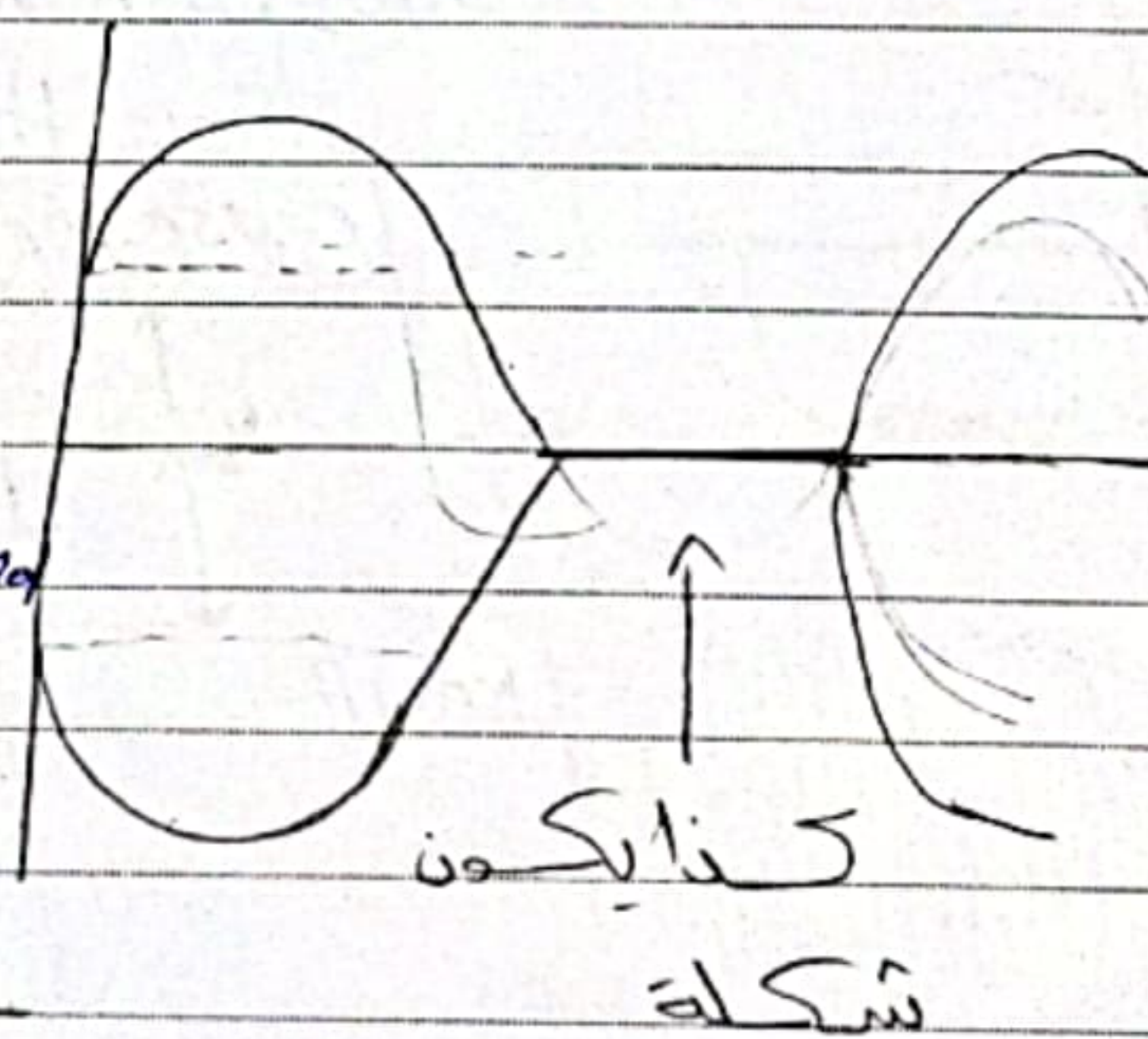
This case is called "over Modulation"

The worst case to send info. cause some of info be lost as shown in figure.



important info:-

When a receiver received this AM signal... it will notice that there's lost in the signal so it will fill this lost by **noise**.



الاستقبال يستلم هذه الإشارة بملء فم
ان جزء من الإشارة مفقود... ولشأن ليسوي
تكمّل هذه الإشارة بملء فم... (noise)
شبهه للفقير ان يتشوش (noise)

Follow.

$$V_{\max} = V_c + V_m \quad (1)$$

$$V_{\min} = V_c - V_m \quad (2)$$

Eq (1) + (2) :-

$$V_{\max} + V_{\min} = 2V_c \Rightarrow V_c = \frac{V_{\max} + V_{\min}}{2}$$

(1) - (2) :-

$$V_{\max} - V_{\min} = 2V_m \Rightarrow V_m = \frac{V_{\max} - V_{\min}}{2}$$

$$m = \frac{V_m}{V_c} \Rightarrow m = \frac{(V_{\max} - V_{\min})/2}{(V_{\max} + V_{\min})/2} \Rightarrow m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

Automatic circuit:-

it's a circuit that prevents over modulation and distortion.

it amplifies the lower level signal and compressing the higher level signal. $\Leftarrow$ This is the main function of Automatic circuit.

Note:-

This circuit is only used in over modulation case to prevent

Follow.

Example 1) sketch the AM freq. spectrum.

$$F_m = 400 \text{ Hz} = 0,4 \text{ KHz}$$

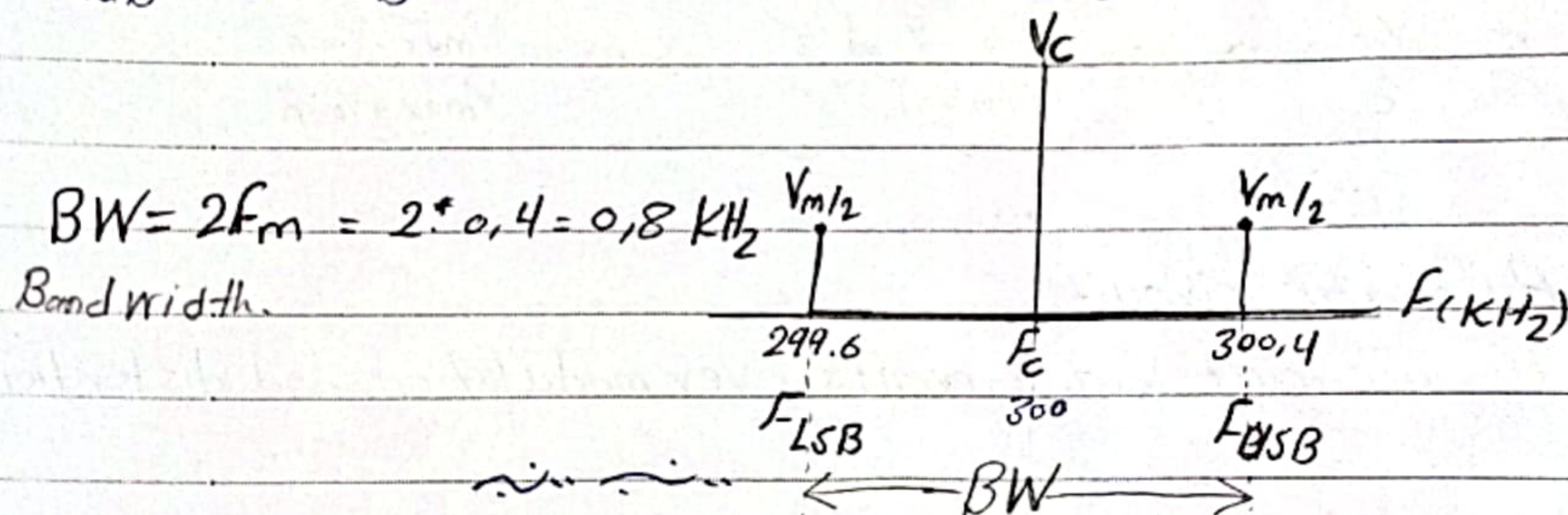
$$F_c = 300 \text{ KHz}$$

Solution.

to sketch the spectrum, we must first find USB, LSB

$$F_{USB} = F_c + F_m = 300 + 0,4 = 300,4 \text{ KHz}$$

$$F_{LSB} = F_c - F_m = 300 - 0,4 = 299,6 \text{ KHz}$$



Example 2) اكتب لي فreq. spectrum
الذي فوق بالضبط
الاضاف فقط القيم.

Example 3)

Given:-

$$V_c = 10 \text{ V} \quad F_c = 1 \text{ MHz} = 10^3 \text{ KHz}$$

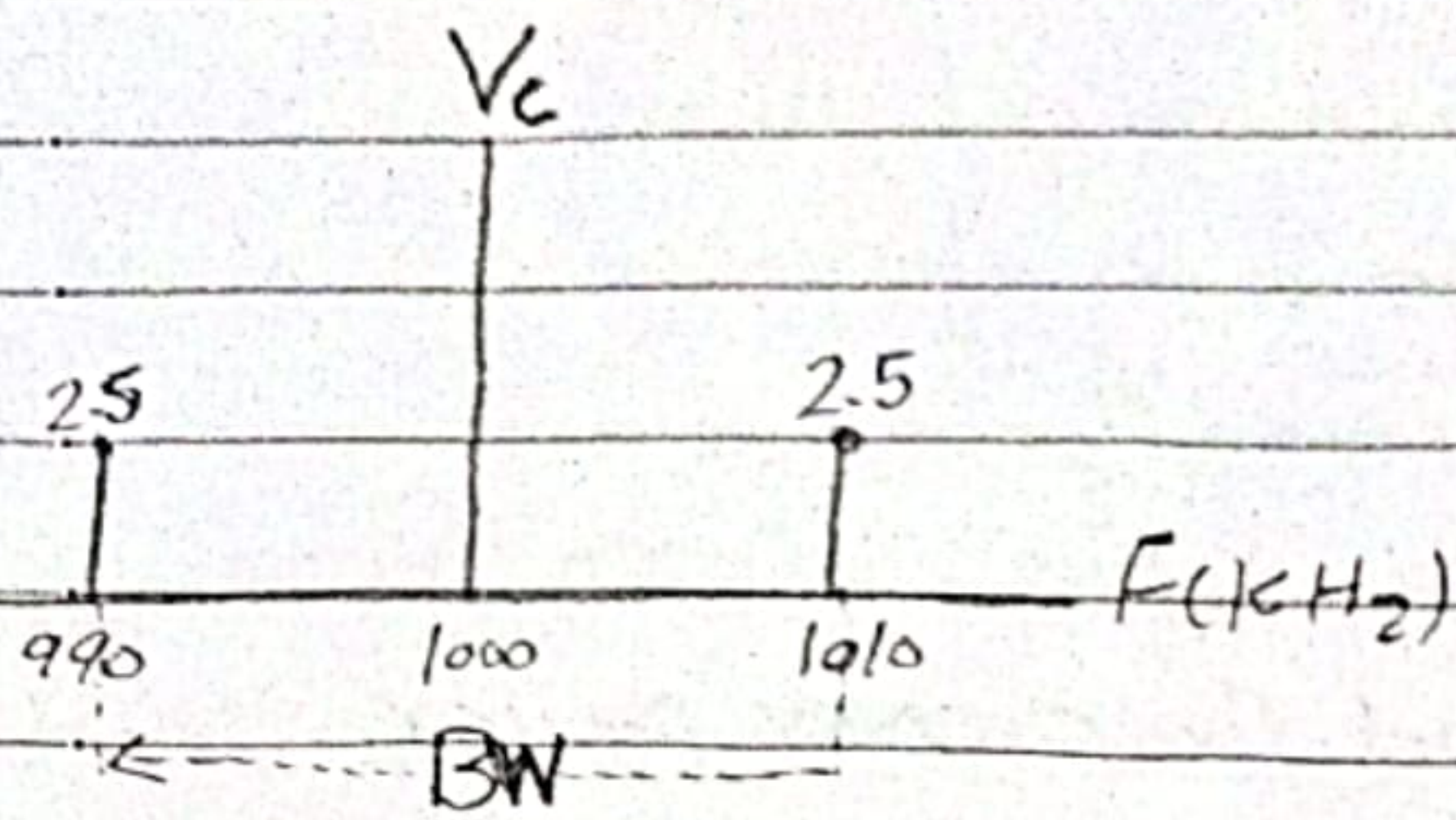
$$V_m = 5 \text{ V} \quad F_m = 10 \text{ KHz} \quad F_{USB} = F_c + F_m = 1010 \text{ KHz}$$

$$F_{LSB} = F_c - F_m = 990 \text{ KHz}$$

$$V_{AM} = V_c \sin 2\pi F_c t + \frac{V_m}{2} \cos 2\pi (F_c - F_m) t - \frac{V_m}{2} \cos 2\pi (F_c + F_m) t$$

$$V_{AM} = 10 \sin 2\pi (10^3) t + \frac{5}{2} \cos 2\pi (1000 - 10) t - \frac{5}{2} \cos 2\pi (1000 + 10) t$$

Follow.



$$BW = 2f_m = 2 \times 10 = 20 \text{ KHz}$$

~~~~~

## AM power.

- Power of AM wave,

$$P = \frac{V_{rms}^2}{R}$$

$$\text{Where } V_{rms} = \frac{V_m}{\sqrt{2}}$$

Carrier Power  $\Rightarrow$ 

$$P_c = \frac{(V_c/\sqrt{2})^2}{R} = \frac{V_c^2}{2R} \quad \leftarrow \text{To find the carrier power}$$

- For one side band,

$$P_{USB} = P_{LSB} = \frac{(V_m/2\sqrt{2})^2}{R} = \frac{V_m^2}{8R}$$

هذا انك لكانت واحد فقط  
لذلك قسمة  $V_m$  على 2

$$P_{USB} = P_{LSB} = \frac{(V_{rms})^2}{R} = \frac{(V_m/2)^2}{R} = \frac{V_m^2}{8R}$$

Where  $V_{rms} = \frac{V_m}{\sqrt{2}}$

هنا بالتفصيل

- For side band power with respect to  $P_c$ :-

$$P_{USB} = P_{LSB} = \frac{m^2}{4} P_c$$

هذا قانون كذا  
انك طالع الحسب بانك اول  
انك طبقه اي بانك ... بالنسبة  
Carrier power.

- For both side band with respect to  $P_c$ :-

$$P_{SB} = \frac{m^2}{2} P_c$$

هذا قانون طبقه لو طالع الحسب  
الـ بانك ... كل (upper, lower)  
upper, lower.

- Total AM power:-

$$P_t = P_c + P_{USB} + P_{LSB}$$

$$P_t = P_c + \frac{m^2}{4} P_c + \frac{m^2}{4} P_c \Rightarrow P_t = P_c \left( 1 + \frac{m^2}{2} \right)$$

Follow.

We care of the maximum power of AM and that occurs

$$\text{When } m=1 \Rightarrow P_t = P_c \left(1 + \frac{1}{2}\right) = 1.5 P_c$$

simultaneous modulation:-

this's mean when there've more than one sine wave to be Modulated

$$m_t^2 = m_1^2 + m_2^2 + m_3^2$$

لا يكون في أكثر من موجة واحدة لأننا إذا Modulated  
نفس الوقت، فهذا، إلكترونيًا.

Example 4)

(under modulation) إذا هذا (under modulation)

Given:-  $P_c = 30 \text{ W}$

$$m = 85\% = \frac{85}{100} = 0.85$$

مطابقة الإشارة  
فقط في غير

required to find:-

a) Total power

b) Power in one sideband.

Solu..

$$a) P_t = P_c \left(1 + \frac{m^2}{2}\right) = 30 \left(1 + \frac{(0.85)^2}{2}\right) = 40.84 \text{ W}$$

تحويل  
مباشرة

$$b) P_{USB} \text{ or } P_{LSB} = \frac{m^2}{4} P_c = \frac{(0.85)^2}{4} \times 30 = 5.4 \text{ W}$$

Ex. 5)

represent  $V_c$

represent  $V_m$

$$V_c = 10 \cos(8\pi \times 10^5 t)$$

$$V_m = 6 \cos(\pi \times 10^5 t)$$

$$V_c = 10 \text{ V}, V_m = 6 \text{ V}, R = 10 \text{ k}\Omega = 10^4 \Omega$$

$$\text{Find) } \omega_c = 8\pi \times 10^5 \quad \omega_m = \pi \times 10^5$$

a) Transmitted total power

b) AM modulated waveform bandwidth.

c) If total power was reduced by 10% determine the value of modulation index

Follow

a)  $P_f = P_c (1 + \frac{m^2}{2})$  we must find  $P_c$  and  $m$ .

$$P_C = \frac{V_C^2}{2R} = \frac{10^2}{2 \times 10^4} = 0.005 \text{ W}$$

$$m = \frac{V_m}{V_c} = \frac{6}{10} = 0.6$$

$$P_f = 0,005 \left( 1 + \frac{(0,6)^2}{2} \right) = 0,0059 \text{ W}$$

ما يقوله السؤال  $\rightarrow$  reduced  
يعني نحسب الجهد من طاقة  
ما يقوله  $\rightarrow$  reduced to .. يعني منه  
بناقة آخر قيمة الذي هو الجهد

b)  $BW = 2f_m$  we must find  $f_m$ .

$$\omega_m = 2\pi f_m \Rightarrow f_m = \frac{\omega_m}{2\pi} = \frac{\pi \times 10^6}{2\pi} = 50 \text{ KHz}$$

$$\therefore BW = 2f_m = 2 \times 50 \text{ K} = 100 \text{ KHz}$$

c) while the reduced power of total power is by 10% that's mean.

$$P_f = 0,0059 \cdot 0,9 = 0,00531 \text{ W}$$

نظریہ یا اقیسہ اساسیہ ال کسان رطل اقیسہ  
لغہ لغہ ماصلة ففقدان

بالتالي،  $\frac{1}{M}$  من هذه  
الفرقة من السؤال  $\frac{1}{M}$  من السؤال  
الذي فيه (m)

$$P_f = P_c \left(1 + \frac{m^2}{2}\right) \quad m?$$

$$\Rightarrow \frac{P_f}{P_c} = 1 + \frac{m^2}{2} \Rightarrow m^2 = 2 \left( \frac{P_f}{P_c} - 1 \right) = 2 \left( \frac{0,00531}{0,005} - 1 \right) = 0,124$$

$m = \sqrt{0,124} = 0,35$  While  $m < 1$  so it's in case of under modulation.

Follow.

Ex. 6)

For peak to peak  $\Rightarrow V_{max(P-P)} = 150V$   $V_{min(P-P)} = 50V$

- we only need 1 peak so  $\therefore V_{max} = \frac{150}{2} = 75V$  ,  $V_{min} = \frac{50}{2} = 25V$

- What's the percentage of total power to unmodulated carrier

$$\text{Power } P_T \Rightarrow \frac{P_T}{P_C}$$

Solution:-

$$P_T = P_C \left(1 + \frac{m^2}{2}\right) \quad \text{Must find } m.$$

$$m = \frac{V_m}{V_c} \quad \text{must find } V_m, V_c$$

$$V_m = \frac{V_{max} - V_{min}}{2} = \frac{75 - 25}{2} = 25V$$

مقسمة على 2  
لأنه 2 قمة  
في 1 دورة

$$V_c = \frac{V_{max} + V_{min}}{2} = \frac{100}{2} = 50V$$

$$m = \frac{V_m}{V_c} = \frac{25}{50} = 0.5$$

$$P_T = P_C \left(1 + \frac{m^2}{2}\right) \Rightarrow \frac{P_T}{P_C} = \left(1 + \frac{(0.5)^2}{2}\right) = 1.125$$

$$\therefore \frac{P_T}{P_C} = 1.125 = 112.5\%$$

Follow.

Ex. 7)

$$V_c = 10 \text{ V}, R = 50 \Omega, m = 50\% = 0,5$$

Find total power and each side band power.

Solu.

a)  $P_f = P_c \left(1 + \frac{m^2}{2}\right)$  must find  $P_c$

$$P_c = \frac{V_c^2}{2R} = \frac{100}{100} = 1 \text{ W}$$

$$P_f = 1 \left(1 + \frac{0,5^2}{2}\right) = 1,125 \text{ W}$$

b)  $P_{USB} = P_{LSB} = \frac{m^2}{4} P_c = \frac{0,25}{4} (1) = 0,0625 \text{ W}$  ✗

8 Ex 8) in this eg. there're two waveforms.

$$m_1 = 40\% = 0,4 \quad I_{f1} = 11 \text{ A}, \quad I_f = 12 \text{ A}$$

- What's the modulation index due to this second wave?

$$\frac{P_f}{P_c} = 1 + \frac{m^2}{2} \quad \text{is relation}$$

$$I_{f1} = I_c \sqrt{1 + \frac{m_1^2}{2}} \Rightarrow I_c = \frac{I_{f1}}{\sqrt{1 + \frac{m_1^2}{2}}} = \frac{11}{\sqrt{1 + \frac{0,4^2}{2}}} = 10,58 \text{ A}$$

$$I_f = I_c \sqrt{1 + \frac{m_2^2}{2}} \Rightarrow \text{cancel } I_c$$

$$\frac{I_f}{I_c} = \sqrt{1 + \frac{m_2^2}{2}} \Rightarrow \left(\frac{I_f}{I_c}\right)^2 = 1 + \frac{m_2^2}{2} \Rightarrow \left(\frac{12}{10,58}\right)^2 = 1 + \frac{m_2^2}{2}$$

$$m_2^2 = 2(1,286 - 1) = 0,572$$

$$m_2 = \sqrt{0,572} = 0,756$$

$$m_4 = \sqrt{m_1^2 + m_2^2} \Rightarrow m_2 = \sqrt{m_4^2 - m_1^2} = \sqrt{0,756^2 - 0,4^2} = 0,642$$

$$P_f = P_c \left(1 + \frac{m^2}{2}\right)$$

$$I_f^2 R = I_c^2 R \left(1 + \frac{m^2}{2}\right)$$

/√

$$I_f = I_c \sqrt{1 + \frac{m^2}{2}}$$

15-7-2021

## Frequency Modulation

FM:-

The frequency of a carrier due to the amplitude of Modu. signal.

Let: the modulating signal:  $V_m = V_m \cos \omega_m t$

A carrier signal:  $V_c = V_c \cos \omega_c t$

→

$$F_{fm} = f_c + K_f V_m$$

$$F_{fm} = f_c + K_f V_m$$

$$F_{fm} = f_c + \delta$$

Where  $F_{fm} \Rightarrow$  final frequency Modulation.

$$\delta = K_f V_m$$

$K_f \Rightarrow$  proportionality constant in Hz/V

$\delta \Rightarrow$  frequency deviation.

→ A maximum freq. deviation occurs when  $\cos \omega_m = 1$

$$F_{fm} = f_c + K_f V_m \leftarrow \text{instantaneous amplitude.}$$

$$= f_c + K_f V_m \cos \omega_m t$$

$$\text{If } t = 0 \Rightarrow \cos \omega_m(0) = 1$$

$$F_{fm} = f_c + K_f V_m$$

$V_{fm} = V_c \cos \phi$  ... ① Where  $V_{fm} \Rightarrow$  signal after Modulation.

$$\phi = \omega_{fm} t$$

المشغلة

$$\phi = \int \omega_{fm} dt$$

$$\text{Where } \omega_{fm} = 2\pi F_{fm}$$

$$F_{fm} = f_c + K_f V_m$$

$$V_m = V_m \cos \omega_m t$$

$$\phi = \int 2\pi [f_c + K_f V_m] dt$$

$$\phi = \int 2\pi f_c dt + \int 2\pi K_f V_m \cos(\omega_m t) dt$$

$$\phi = 2\pi f_c t + \frac{2\pi K_f V_m \sin 2\pi f_m t}{2\pi f_m}$$

$$\phi = \omega_c t + m_f \sin \omega_m t$$

$$m_f = \frac{K_f V_m}{f_m} \leftarrow \text{modulation index}$$

Follow.

$\phi = \omega_c t + m_f \sin \omega_m t$  by substitute it in ①.

$$V_m = V_c \cos \phi$$

$$V_m = V_c \cos [\omega_c t + m_f \sin \omega_m t]$$

$m_f$  is the modulation index of FM and it's equal:-

$$m_f = \frac{k_f V_m}{f_m} = \frac{\delta}{f_m}$$

$$\delta = m_f f_m$$

Conclusion of this proved is:-

$$\delta [\text{deviation freq.}] = k_f V_m \cos \phi$$

is the maximum FM Freq. from the carrier Freq.

$m_f$  [Modulation Index]:

is the ratio of the deviation Freq. to the modu. Freq.

$$m_f = \frac{\delta}{f_m} = \frac{k_f V_m}{f_m}$$

Amplitude of Modu. signal.  
Freq. of modulating signal.

Note:-

- A modulation index is directly proportion to the Amplitude of modulating signal and inversely modulating signal Frequency.

- A deviation Freq. is depend on the Amplitude of Modulating signal.

$$\delta = k_f V_m$$

Note that:-

$$m_f = \frac{\delta}{f_m} \Rightarrow \delta = m_f f_m$$

$$\therefore \delta = k_f V_m$$

$$\Rightarrow k_f V_m = m_f f_m$$

proven

Ex. 10) What's the new  $\delta$  if the  $V_m = 7.2V$ ?

Given:-

$$f_m = 500 \text{ kHz} \quad V_m = 2.4 \text{ V} \quad \delta = 4.8 \text{ kHz} \quad V_{m_{\text{new}}} = 7.2$$

Find the new  $\delta$  and  $m_f$  in each case.

Solu..

$$m_f = \frac{\delta}{f_m} = \frac{4.8 \text{ K}}{0.5 \text{ K}} = 9.6$$

$$\delta = k_f V_m$$

$$k_f = \frac{\delta}{V_m} = \frac{4.8 \text{ K}}{2.4} = 2 \text{ kHz/V}$$

$$\text{For } V_{m_{\text{new}}} = 7.2 \text{ V}$$

$$\delta_{\text{new}} = k_f V_{m_{\text{new}}}$$

$$= 2 \times 7.2 = 14.4 \text{ kHz}$$

هذا يدل انه مقدار الانحراف يساوي 2 كيلو هيرتز لكل فولت.  
deviation

وهذا يظل ثابتا في حالة كمال لجميع السيجنل.

$$m_{f_{\text{new}}} = \frac{\delta_{\text{new}}}{f_m} = \frac{14.4 \text{ K}}{0.5 \text{ K}} = 28.8$$

$$\text{Ex. 11) } v_{f_m} = 12 \cos[6 \times 10^8 + 5 \sin 1250t]$$

$$\text{Given :- } V_c = 12 \text{ V} \quad \omega_c = 6 \times 10^8 \text{ rad/s} \quad \omega_m = 1250 \text{ rad/s} \quad m_f = 5$$

Find:-

$$R = 10 \Omega$$

$f_c, \delta$  What power will this FM dissipate in  $10 \Omega$ ?

$$\omega_c = 2\pi f_c \Rightarrow f_c = \frac{\omega_c}{2\pi} = \frac{6 \times 10^8}{2\pi} = 95.5 \text{ MHz}$$

$$\omega_m = 2\pi f_m \Rightarrow f_m = \frac{\omega_m}{2\pi} = \frac{1250}{2\pi} = 198.9 \text{ Hz}$$

$$\delta = f_m \cdot m_f = 198.9 \times 5 = 994.5 \text{ Hz}$$

$$P_c = \frac{V_c^2}{2R} = \frac{12^2}{20} = 7.2 \text{ W}$$

## 17. Frequency Modulation (FM) and Amplitude Modulation (AM)

→  $K_f$ :

is selecting how much freq. can be change per one Volt.

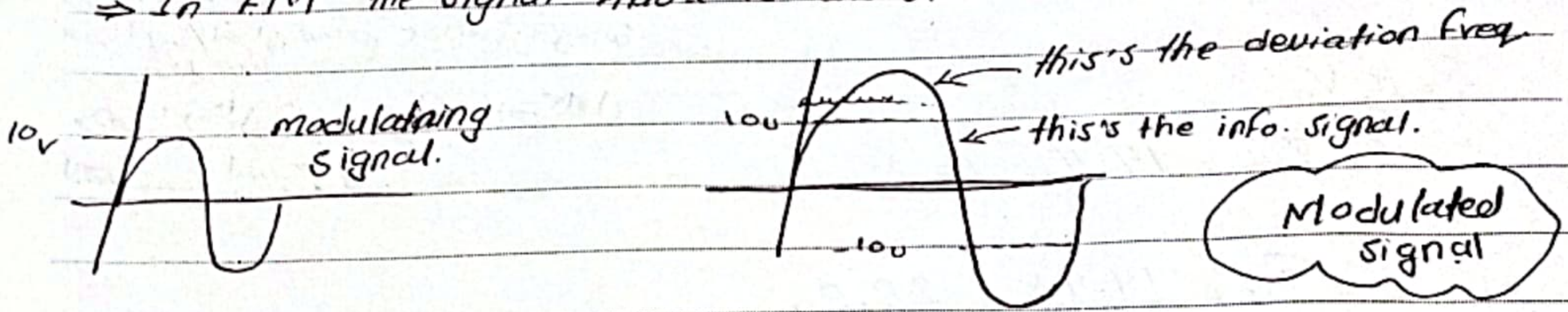
→  $m_f$ :

its value must be bigger than 1, cause the amplitude of signal multiplying by  $K_f$  will be bigger than the value of freq.

$$\text{Modulating signal } f_m \Rightarrow m_f = \frac{K_f V_m}{f_m}$$

AM:  $\Delta f \propto V_m$

→ In FM the signal info. is exists in a carrier it self



→ FM used to send voice, While AM for photo & video.

→ The deviation is independent of signal freq.